

```
build.gradle docker-compose.yml Dockerfile gradle gradlew
gradlew.bat LICENSE.txt mvnw mvnw.cmd pom.xml readme.md
settings.gradle src target
osfy@ubuntu-22-04-1-lts:~/Desktop/spring-petclinic$ docker
image build -t petclinic:latest .
```

The output screen will show you the successful execution of the command.

```
Successfully built 6f78e1ae176e
Successfully tagged petclinic:latest
```

Now verify that the PetClinic image is available while you execute the *docker images* command.

```
osfy@ubuntu-22-04-1-lts:~/Desktop/spring-petclinic$ docker
images
```

REPOSITORY	SIZE	TAG	IMAGE ID
petclinic		latest	
6f78e1ae176e	6 minutes ago	707MB	
registry.k8s.io/kube-apiserver		v1.25.3	
0346dbd74bcb	7 weeks ago	128MB	
registry.k8s.io/kube-scheduler		v1.25.3	
6d23ec0e8b87	7 weeks ago	50.6MB	
registry.k8s.io/kube-controller-manager		v1.25.3	
603999231275	7 weeks ago	117MB	
registry.k8s.io/kube-proxy		v1.25.3	
beaaf00edd38	7 weeks ago	61.7MB	
openjdk		11	
47a932d998b7	4 months ago	654MB	
registry.k8s.io/pause		3.8	
4873874c08ef	5 months ago	711kB	
registry.k8s.io/etcd		3.5.4-0	
a8a176a5d5d6	6 months ago	300MB	
registry.k8s.io/coredns/coredns		v1.9.3	
5185b96f0bec	6 months ago	48.8MB	
k8s.gcr.io/pause		3.6	
6270bb605e12	15 months ago	683kB	
gcr.io/k8s-minikube/storage-provisioner		v5	
6e38f40d628d	20 months ago	31.5MB	

We will use the Docker images available in the local environment and not from the public or private Docker registry. So set *imagePullPolicy* value to *Never* in the YAML file. Here is the YAML file (for deployment and service) to deploy a sample application in the minikube cluster:

```
apiVersion: apps/v1
kind: Deployment
metadata:
```

```
creationTimestamp: null
labels:
  app: petclinic
name: petclinic
spec:
  replicas: 1
  selector:
    matchLabels:
      app: petclinic
  strategy: {}
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: petclinic
    spec:
      containers:
        - image: petclinic:latest
          name: petclinic
          resources: {}
          imagePullPolicy: Never
status: {}
---
apiVersion: v1
kind: Service
metadata:
  name: petclinic
spec:
  type: LoadBalancer
  selector:
    app: petclinic
  ports:
    - protocol: TCP
      port: 8080
      targetPort: 8080
```

Use the *kubectl apply* command to create service and deployment available in the YAML file.

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ kubectl apply -f
petclinic-deployment.yaml
deployment.apps/petclinic created
service/petclinic created
```

The minikube tunnel runs as a process. It helps to expose the Load Balancer Service type. We need to run the minikube *tunnel* command in a separate terminal so the load balancer continues to run until we press CTRL + C. It helps to create a network route on the host. Open a new terminal window or a tab and execute the *minikube tunnel* command:

```
osfy@ubuntu-22-04-1-lts:~/Desktop$ minikube tunnel
```